

TRADING SURVEILLANCE SYSTEM

SUSPICIOUS ACTIVITY DETECTION REPORT

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6600	600	1.0	1.0
TOTAL TRADES	SUSPICIOUS	ROC-AUC	F1-SCORE

EXECUTIVE SUMMARY

This report presents the findings of the ML-powered trading surveillance system. Out of **6,600** trades analyzed, **600** (9.1%) were flagged as suspicious. The model achieved a ROC-AUC of **1.0** with precision **1.0** and recall **1.0**.

The following manipulation patterns were detected and are detailed in subsequent sections: **Pump & Dump, Spoofing, Layering, Wash Trading, and Front Running.**

MODEL PERFORMANCE METRICS

METRIC	VALUE	ASSESSMENT
ROC-AUC Score	1.0	EXCELLENT
Average Precision	1.0	EXCELLENT
Precision (Susp.)	1.0	EXCELLENT
Recall (Susp.)	1.0	EXCELLENT
F1-Score (Susp.)	1.0	EXCELLENT
Accuracy	1.0	EXCELLENT
CV AUC Mean	1.0	EXCELLENT
CV AUC Std	0.0	FAIR

MODELS USED IN ENSEMBLE

- RandomForest(n=300)
- GradientBoosting(n=200)
- IsolationForest(n=250)
- LocalOutlierFactor(k=25)

TOP FLAGGED TRADES

The following 20 trades were identified as highest risk by the ML ensemble. Each entry includes the primary SHAP feature attributions explaining the detection.

#	TRADE ID	RISK SCORE	LEVEL	PATTERN	TOP REASON
1	TRD002531	0.7471	HIGH	Layering	freq_counterparty_ratio (suspici
2	TRD001784	0.7303	HIGH	Layering	freq_counterparty_ratio (suspici
3	TRD000581	0.7241	HIGH	Pump & Dump	freq_counterparty_ratio (suspici
4	TRD001447	0.7203	HIGH	Layering	freq_counterparty_ratio (suspici
5	TRD001079	0.7199	HIGH	Layering	freq_counterparty_ratio (suspici
6	TRD000796	0.7186	HIGH	Layering	freq_counterparty_ratio (suspici
7	TRD001818	0.7186	HIGH	Pump & Dump	market_dominance (suspicious)
8	TRD002887	0.7177	HIGH	Pump & Dump	freq_counterparty_ratio (suspici
9	TRD005475	0.7166	HIGH	Pump & Dump	market_dominance (suspicious)
10	TRD003883	0.7157	HIGH	Pump & Dump	volume_to_market_ratio (suspicio
11	TRD003972	0.7130	HIGH	Pump & Dump	market_dominance (suspicious)
12	TRD002916	0.7127	HIGH	Layering	freq_counterparty_ratio (suspici
13	TRD002594	0.7124	HIGH	Layering	freq_counterparty_ratio (suspici
14	TRD004884	0.7111	HIGH	Pump & Dump	freq_counterparty_ratio (suspici
15	TRD005832	0.7107	HIGH	Pump & Dump	freq_counterparty_ratio (suspici

SUSPICIOUS PATTERN ANALYSIS

Pump & Dump

Artificial price inflation through coordinated buying, followed by rapid selling at peak price.

Spoofing

Placing large orders with no intention to execute, creating false market depth to mislead traders.

Layering

Multiple orders at different price levels, cancelled immediately after affecting other market participants.

Wash Trading

Trading with oneself to create artificial volume and price movement without genuine market participation.

Front Running

Trading ahead of known pending large client orders to profit from the anticipated price movement.

REGULATORY RECOMMENDATIONS

PRIORITY	ACTION	DESCRIPTION
CRITICAL	Immediate Review	All HIGH risk trades (score >= 0.70) should be reviewed by compliance officers within 24 hours.
HIGH	Pattern Monitoring	Traders with repeated spoofing or layering signals should be placed on enhanced monitoring watchlist.
HIGH	Regulatory Filing	Trades matching wash trading criteria should be reported to the relevant regulatory authority per applicable rules.
MEDIUM	Model Retraining	The ML model should be retrained monthly with newly labeled data to maintain detection accuracy.
MEDIUM	Alert Thresholds	Consider lowering the review threshold to 0.35 during periods of elevated market volatility.
MEDIUM	Network Analysis	Review trader network clusters flagged as coordinated manipulation rings for potential collusion.

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